## Uses

***Smartphones*** *run on rechargeable batteries that use lithium to store electricity.*

Smartphone

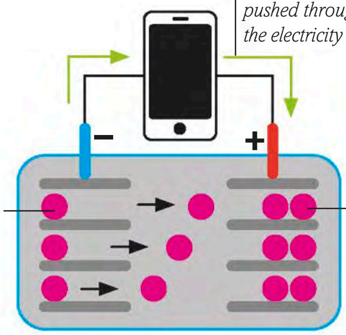
Hale telescope mirror

***Lithium added to the glass in this mirror*** *stops the disc warping at extreme temperatures.*

## LITHIUM-ION BATTERY

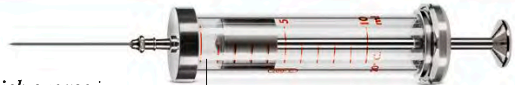
Lithium-ion batteries are widely used in digital devices. They store electrical energy to power gadgets and are rechargeable. This diagram shows a device's battery in use; when it is charging, this process is reversed.

*1. Inside the battery, positively charged lithium ions move from the negative electrode (-) to the positive electrode (+).*




*2. The positive electrode receives lithium ions as the battery loses charge.*

*3. As ions move inside the battery, negatively charged electrons are pushed through the phone, providing the electricity to make it work.*



Syringe

***Lithium coating on the inside of some syringes*** *delays the clotting of the blood sample.*

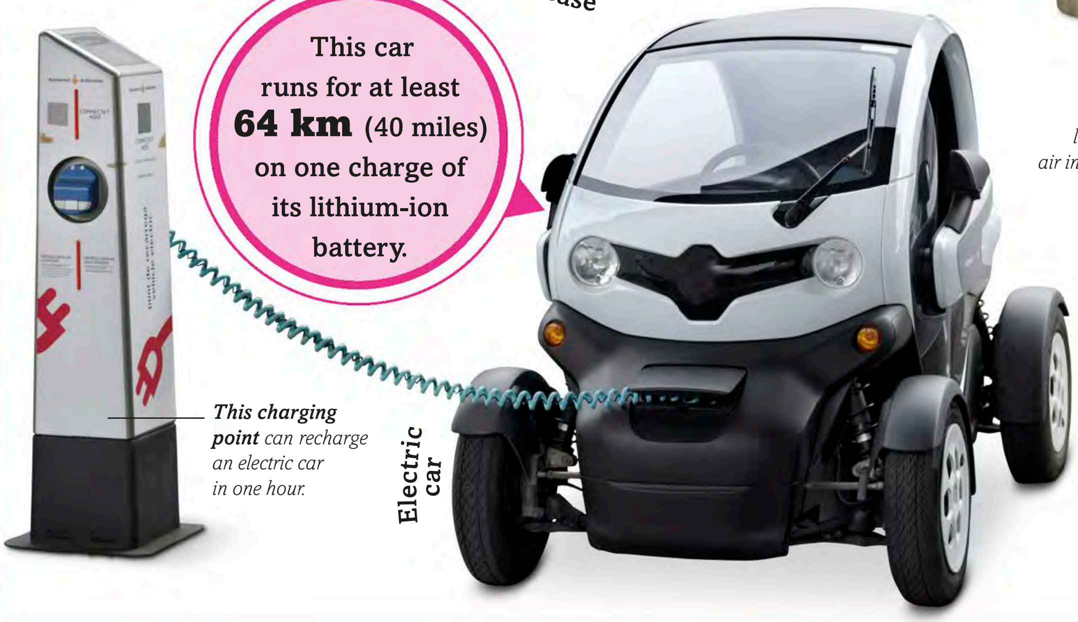
***Lithium-rich grease*** *is used to keep mechanical parts of engines running smoothly, even when hot.*


Grease

***Some artificial teeth*** *contain lithium disclate, which makes them strong.*

Artificial teeth

This car runs for at least **64 km** (40 miles) on one charge of its lithium-ion battery.



***This charging point*** *can recharge an electric car in one hour.*

Electric car

***This air scrubber*** *used lithium hydroxide to purify the air inside the Apollo 13 spacecraft.*

Air scrubber

**telescopes**. The main use for lithium is in rechargeable batteries. Lithium-ion batteries are small but powerful, so they are ideal for **smartphones** and tablet computers. Larger lithium batteries can power **electric cars**, which are less polluting than petrol-powered ones. A soapy compound called lithium stearate is used to make **grease**, which helps automobile engines run smoothly. This element also forms hard ceramics that are used to produce strong **artificial teeth**. Lithium compounds are used in some medicines as well.